



DEFENCE ELECTRONICS

cal boundaries of the material. Some of the commonly used photoemissive photosensors include vacuum photodiodes, photomultiplier tubes and image-intensifier tubes.

Photomultiplier tubes are extremely sensitive photosensors operating in the ultraviolet, visible and near-infrared spectrum. These have internal gain of the order of 10^6 and can detect even a single photon of light. They are constructed from a glass vacuum tube that houses a photocathode, several dynodes and an anode. When incident photons strike the photocathode, electrons are produced as a result of the photoelectric effect. These electrons accelerate towards the anode and in the process, electron multiplication takes place due to secondary emission from dynodes.

Salient features of photomultiplier tubes include low noise, high-frequency response and large active area. By virtue of these features, photomultiplier tubes are used in nuclear and particle physics,



Fig. 11: Representative photomultiplier tube

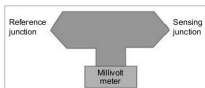


Fig. 12: Thermocouple having two junctions

astronomy, medical imaging and motion-picture film scanning. Fig. 11 shows a representative photomultiplier tube.

Thermal sensors

Thermal sensors absorb radiation, which causes a change in temperature and hence physical or electrical property of the sensor. In other words, thermal sensors respond to

any change in their bulk temperature caused by the incident radiation. Thermocouple, thermopile, bolometer and pyroelectric sensors are thermal sensors. Thermal sensors lack the sensitivity of photoelectric sensors and are generally slower in response, but these have a wider spectral response. Most of these sensors are passive devices, requiring no bias.

Thermocouples and thermopiles

Thermocouple sensors are based on Seebeck effect, that is, temperature change at the junction of two dissimilar metals generates a proportionate EMF. Commonly used thermocouple materials are bismuth-antimony, iron-constantan and copper-constantan. Their temperature coefficients are $100\mu\text{V}/^\circ\text{C}$, $54\mu\text{V}/^\circ\text{C}$ and $39\mu\text{V}/^\circ\text{C}$, respectively. To compensate for changes in the ambient temperature, thermocouples generally have two junctions: measuring junction and reference junction (Fig. 12).

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